Calculus 3

Thi Hoai Thuong NGUYEN

University of Science, VNU-HCM, Vietnam

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Outline

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We start with a plane curve C given by the parametric equations

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases} \qquad \text{with } a \leq t \leq b$$

- The parametric equations of the line through the points $A(x_A, y_A)$ and $B(x_B, y_B)$:

$$\begin{cases} x = x_A + (x_B - x_A)t \\ y = y_A + (y_B - y_A)t \end{cases} \qquad \text{with } 0 \leq t \leq 1$$

- The parametric equations of the circle $(x - a)^2 + (y - b)^2 = r^2$:

$$\begin{cases} x = a + r\cos(t) \\ y = b + r\sin(t) \end{cases} \qquad \text{with } 0 \leq t \leq 2\pi$$

- The parametric equations of the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$:

$$\begin{cases} x = a\cos(t) \\ y = b\sin(t) \end{cases} \qquad \text{with } 0 \leq t \leq 2\pi.$$

The parametric equations of the curve

- The parametric equations of the curve (C) in the polar coordinates:

$$\begin{cases} x = r(\varphi)\cos(\varphi) \\ y = r(\varphi)\sin(\varphi) \end{cases} \qquad \text{with } \alpha \leq \varphi \leq \beta.$$

- The parametric equations of the smooth space curve (C):

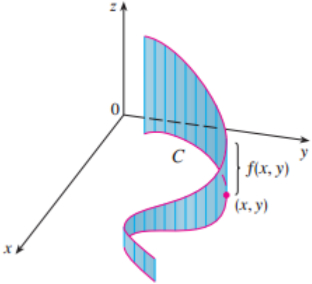
$$\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \qquad \text{with } a \leq t \leq b.$$

Example: Find the parametric equations of the intersection of the cylinder $x^2 + y^2 = 4$ and the plane $z = 3$.

Example: Find the parametric equations of the intersection of the sphere $x^2 + y^2 + z^2 = 6z$ and the plane $z = 3 - x$.

The area of the "fence" or "curtain"

If $f(x,y) \geq 0$ and C is the curve in the plane Oxy, $\int_C f(x,y)ds$ represents the area of one side of the "fence" or "curtain", whose base is C and whose height above the point (x,y) is $f(x,y)$.

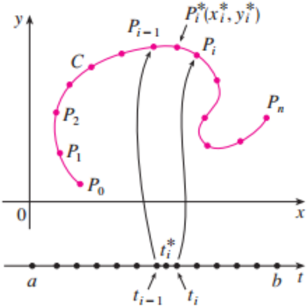


Line integral of type 1

We start with a plane curve C given by the parametric equations

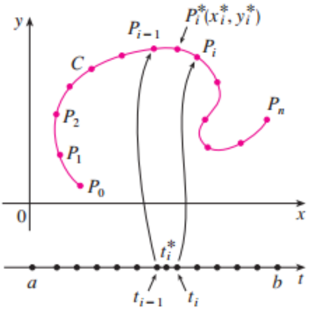
$$x = x(t), \quad y = y(t), \quad a \leq t \leq b$$

and we assume that C is a smooth curve.



If we divide the parameter interval $[a, b]$ into n sub-intervals $[t_{i-1}, t_i]$ of equal width. and we let $x_i = x(t_i)$ and $y_i = y(t_i)$, then the corresponding points $P_i(x_i, y_i)$ divide C into n subarcs with lengths $\Delta s_1, \Delta s_2, ..., \Delta s_n$.

Line integral of type 1



We choose any point $P_i^*(x_i^*, y_i^*)$ in the ith subarc. (This corresponds to a point t_i^* in $[t_{i-1}, t_i]$).
Now if f is any function of two variables whose domain includes the curve C, we evaluate f at the point (x_i^*, y_i^*), multiply by the length Δs_i of the subarc, and form the sum

$$\sum_{i=1}^{n} f(x_i^*, y_i^*)\Delta s_i$$

Line integral of type 1

Then we take the limit of these sums and make the following definition by analogy with a single integral

Definition

If f is defned on a smooth curve C given by

$$x = x(t), \quad y = y(t), \quad a \leq t \leq b$$

then the line integral of f along C is

$$\int_C f(x, y) ds = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i^*, y_i^*) \Delta s_i$$

if this limit exists.

The following formula can be used to evaluate the line integral:

$$\int_C f(x, y) ds = \int_a^b f\left(x(t), y(t)\right) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt. \tag{1}$$

<u>Remark</u>: The value of the line integral **does not depend on** the parametrization of the curve, provided that the curve is traversed exactly once as t increases from a to b.

Line integral of type 1

Example: Evaluate $\int_C (2 + x^2 y)\, ds$, where C is the upper half of the unit circle $x^2 + y^2 = 1$.

Property of the line integral of type 1

Property

Suppose now that C is a piecewise-smooth curve and C is a union of a finite number of smooth curves $C_1, C_2, ..., C_n$. Then we define the integral of f along C as the sum of the integrals of f along each of the smooth pieces of C

$$\int_C f(x,y)ds = \int_{C_1} f(x,y)ds + \int_{C_2} f(x,y)ds + ... + \int_{C_n} f(x,y)ds.$$

Property of the line integral of type 1

Example: Evaluate $\int_C 2x\, ds$, where C consists of the arc C_1 of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$ followed by the vertical line segment C_2 from $(1, 1)$ to $(1, 2)$.

Line Integrals in Space

Definition

If f is a function of three variables that is continuous on some region containing C given by

$$x = x(t), \quad y = y(t), \quad z = z(t), \quad a \leq t \leq b$$

then the line integral of f along C is

$$\int_C f(x,y,z)ds = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*, y_i^*, z_i^*\right) \Delta s_i$$

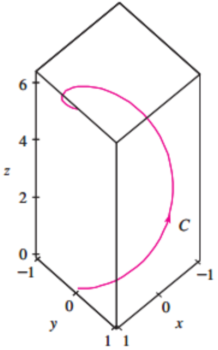
if this limit exists.

The following formula can be used to evaluate the line integral:

$$\int_C f(x,y,z)ds = \int_a^b f\left(x(t), y(t), z(t)\right) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt. \tag{2}$$

Example: Evaluate $\int_C y\sin(z)ds$, where C is the circular helix given by the equations

$$\begin{cases} x = \cos(t) \\ y = \sin(t) \\ z = t \end{cases} \qquad \text{with } 0 \le t \le 2\pi.$$



Line integral of type 2

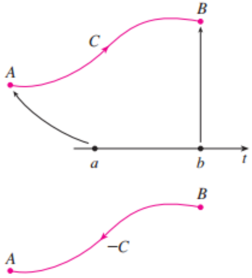
Definition

The line integral of type 2 of the function $F(x, y) = (P(x, y), Q(x, y))$ over a smooth curve C is

$$\int_C F(x, y)ds = \int_C P(x, y)dx + Q(x, y)dy.$$

The properties of Line integral of type 2

1) If $-C$ denotes the curve consisting of the same points as C but with the opposite orientation



then

$$\int_{-C} F(x, y)ds = -\int_{C} F(x, y)ds$$

2) If $C = C_1 \cup C_2$ then

$$\int_{C} F(x, y) = \int_{C_1} F(x, y)ds + \int_{C_2} F(x, y)ds.$$

Line integral of type 2

Evaluate the Line integral of type 2

Let the curve C is given by the parametric equation

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases} \qquad a \leq t \leq b$$

Then,

$$\begin{aligned} \int_C F(x,y)ds &= \int_C P(x,y)dx + Q(x,y)dy \\ &= \int_a^b \left[P\big(x(t),y(t)\big)x'(t) + Q\big(x(t),y(t)\big)y'(t)\right]dt. \end{aligned}$$

Example: Evaluate $I = \int_C -ydx + xdy$, where C is the half circle $x^2 + y^2 = 4, x \geq 0$ from $A(0, -2)$ to $B(0, 2)$.

Line integral of type 2

Evaluate the Line integral of type 2 in space

Let the curve C is given by the parametric equation

$$\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \qquad a \leq t \leq b.$$

Then,

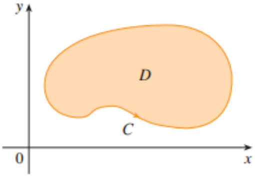
$$\begin{aligned} &\int_C F(x,y,z)ds \\ &= \int_C P(x,y,z)dx + Q(x,y,z)dy + R(x,y,z)dz \\ &= \int_a^b \left[P\big(x(t),y(t),z(t)\big)x'(t) + Q\big(x(t),y(t),z(t)\big)y'(t) + R\big(x(t),y(t),z(t)\big)z'(t)\right]dt. \end{aligned}$$

Line integral of type 2

Example: Evaluate $\int_C y\,dx + z\,dy + x\,dz$, where C consists of the line segment C_1 from $(2, 0, 0)$ to $(3, 4, 5)$, followed by the vertical line segment C_2 from $(3, 4, 5)$ to $(3, 4, 0)$.

Green's theorem

Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C.



In stating Green's Theorem, we use the convention that the positive orientation of a simple closed curve C is the direction in which the region Dis always on the left as the point traverses C.

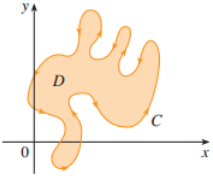


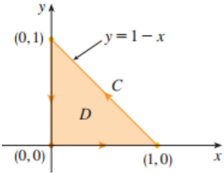
Figure: Positive orientation

Green's theorem

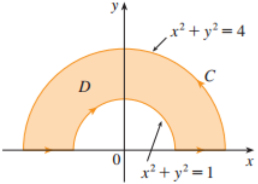
Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C P(x,y)dx + Q(x,y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Example: Evaluate $\int_C x^4 dx + xy\, dy$, where C is the triangular curve consisting of the line segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.



Example: Evaluate $\int_C y^2 dx + 3xy\, dy$, where C is the boundary of the semiannular region Din the upper half-plane between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.



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